Cambridge (CIE) A Level Maths: Pure 1

Your notes

Arithmetic Progressions

Contents

- Arithmetic Progressions
- Sum of Arithmetic Progressions

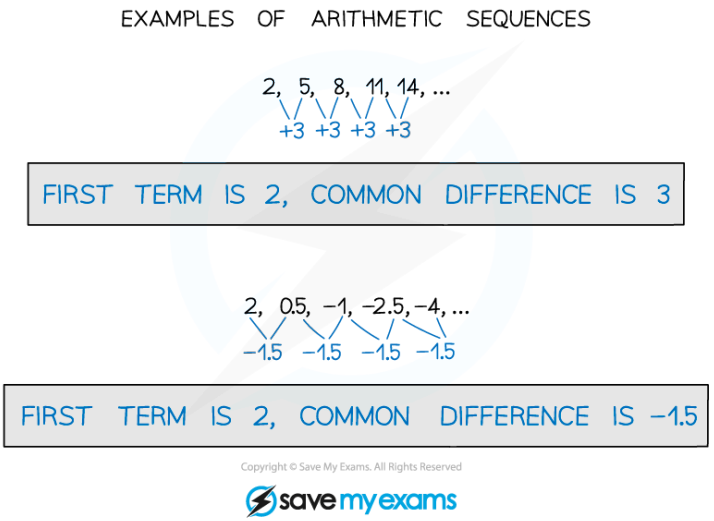
© 2025 Save My Exams, Ltd.

Your notes

Arithmetic progressions

What are arithmetic progressions?

- In an **arithmetic progression** (also called **arithmetic sequence**), the difference between consecutive terms in the sequence is constant



- That constant difference is known as the **common difference** of the sequence
- You need to know the ***n*th term formula** for an arithmetic progression

$$u_n = a + (n-1)d$$

  - a is the first term
  - d is the common difference

Examiner Tips and Tricks

- The formula is given in the formula booklet
- If you know two terms in an arithmetic sequence you can find ***a*** and ***d*** using simultaneous equations.

e.g. THE 15th TERM OF AN ARITHMETIC SEQUENCE IS −19.
THE 21st TERM IS −28.
FIND THE FIRST TERM AND THE COMMON DIFFERENCE.

$u_n = a + (n-1)d$

$a + 14d = -19$ (1)

$a + 20d = -28$ (2)

SOLVE THESE SIMULTANEOUS EQUATIONS

$6d = -9$

$d = -1.5$

(2) MINUS (1)

$a + 14(-1.5) = -19$

$a = 2$

SUBSTITUTE INTO (1)

THE FIRST TERM IS 2 AND
THE COMMON DIFFERENCE IS −1.5

Copyright © Save My Exams. All Rights Reserved
save my exams

Worked Example

The first three terms in an arithmetic sequence are $12, k^2, 5k, \ldots$

Find two possible values of k.

THIS IS AN ARITHMETIC SEQUENCE SO THERE'S A SINGLE COMMON DIFFERENCE

$d = k^2 - 12$ — 2nd TERM MINUS 1st TERM

AND

$d = 5k - k^2$ — 3rd TERM MINUS 2nd TERM

SO $k^2 - 12 = 5k - k^2$

$2k^2 - 5k - 12 = 0$

$(2k+3)(k-4) = 0$ — SOLVE THE QUADRATIC

$k = -\frac{3}{2}$ OR $k = 4$

Copyright © Save My Exams. All Rights Reserved
save my exams

Your notes

Your notes

Sum of arithmetic progressions

How do I find the sum of an arithmetic progression?

- The **sum** of the terms of an **arithmetic progression** is sometimes called an **arithmetic series**

1, 3, 5, 7, 9, ... IS AN ARITHMETIC SEQUENCE

1 + 3 + 5 + 7 + 9 + ... IS AN ARITHMETIC SERIES

Copyright © Save My Exams. All Rights Reserved

save my exams

- The following formulae will let you find the sum of the first ***n*** terms of an arithmetic progression:

$$S_n = \frac{n}{2}(2a + (n-1)d) \text{ or } S_n = \frac{n}{2}(a + l)$$

  - *a* is the first term
  - *d* is the common difference
  - *l* is the last term
- You can use whichever formula is more convenient for a given question
- The ***a*** and the ***d*** in those formulae are exactly the same as the ones used with **arithmetic progressions**

How do I derive the formula for the sum of an arithmetic progression?

- Learn this proof of the sum of an arithmetic progression formula – you can be asked to give it on the exam:
  - Write the terms out once in order
  - Write the terms out again in reverse order
  - Add the two sums together
    - The terms will pair up to give the same sum $2a + (n-1)d$
    - There will be *n* of these terms
  - Divide by two as two of the sums have been added together

Your notes

THE SUM OF THE FIRST n TERMS IS

$$S_n = a + (a+d) + (a+2d) + \ldots + (a+(n-2)d) + (a+(n-1)d) \qquad (1)$$

NOW WRITE THE SAME SUM THE OTHER WAY ROUND

$$S_n = (a+(n-1)d) + (a+(n-2)d) + \ldots + (a+2d) + (a+d) + a \qquad (2)$$

ADD (1) AND (2) TOGETHER TERM BY TERM

$$2S_n = n(2a+(n-1)d)$$

EACH OF THE SUMS CONTAINS n TERMS

$a+(a+(n-1)d) = 2a+(n-1)d$
$(a+d)+(a+(n-2)d) = 2a+(n-1)d$
etc.

DIVIDING BY 2 GIVES

$$S_n = \frac{n}{2}(2a+(n-1)d)$$

Copyright © Save My Exams. All Rights Reserved
save my exams

Examiner Tips and Tricks

The arithmetic series formulae are in the formulae booklet – you don't need to memorise them.

Worked Example

The ninth term of an arithmetic progression is 28. The fifteenth term is 46. The sum of the first n terms is 2500.

a) Show that $3n^2 + 5n - 5000 = 0$.

b) Hence find the value of n.

a) $a + (9 - 1)d = a + 8d = 28$ (1)

$a + (15 - 1)d = a + 14d = 46$ (2)

SIMULTANEOUS EQUATIONS

$6d = 18$

$d = 3$

(2) MINUS (1)

$a = 28 - 8(3)$

$a = 4$

SUBSTITUTE INTO (1)

SO $\frac{n}{2}(2(4) + (n - 1)(3)) = 2500$

S_n FORMULA

$n(3n + 5) = 5000$

$3n^2 + 5n - 5000 = 0$

b) $(3n + 125)(n - 40) = 0$

SOLVE THE QUADRATIC – YOU COULD ALSO USE THE QUADRATIC FORMULA

$n = -\frac{125}{3}$ OR $n = 40$

BUT n MUST BE A POSITIVE INTEGER

$n = 40$

Copyright © Save My Exams. All Rights Reserved

Your notes